

A Practical Guide to Adversarial machine learning

TOPIC -- ADVERSARIAL MACHINE LEARNING

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adversarial training of a linear regression model with input perturbations restricted by the infinity-norm closely resembles Lasso regression, and that adversarial training of a linear regression model with input perturbations restricted by the 2-norm closely resembles Ridge regression.

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History At the MIT Spam Conference in January 2004, John Graham-Cumming showed that a machine-learning spam filter could be used to defeat another machine-learning spam filter by automatically learning which words to add to a spam email to get the email classified as not spam. In 2004, Nilesch Dalvi and others noted that linear classifiers used in spam filters could be defeated by simple "evasion attacks" as spammers inserted "good words" into their spam emails. (Around 2007, some spammers added random noise to fuzz words within "image spam" in order to defeat OCR-based filters.) In 2006, Marco Barreno and others published "Can Machine Learning Be Secure?", outlining a broad taxonomy of attacks. As late as 2013 many researchers continued to hope that non-linear classifiers (such as support vector machines and neural networks) might be robust to adversaries, until Battista Biggio and others demonstrated the first gradient-based attacks on such machine-learning models (2012–2013). In 2012, deep neural networks began to dominate computer vision problems; starting in 2014, Christian Szegedy and others demonstrated that deep neural networks could be fooled by adversaries, again using a gradient-based attack to craft adversarial perturbations. Further work would show that adversarial attacks are harder to produce in uncontrolled environments, due to the different environmental constraints that cancel out the effect of noise. For example, any small rotation or slight illumination on an adversarial image can destroy the adversariality. In addition, researchers such as Google Brain's Nick Frosst point out that it is much easier to make self-driving cars miss stop signs by physically removing the sign itself, rather than creating adversarial examples. Frosst also believes that the adversarial machine learning community incorrectly assumes models trained on a certain data distribution will also perform well on a completely different data distribution. He suggests that a new approach to machine learning should be explored, and is currently working on a unique neural network that has characteristics more similar to human

perception than state-of-the-art approaches. While adversarial machine learning continues to be heavily rooted in academia, large tech companies such as Google, Microsoft, and IBM have begun curating documentation and open source code bases to allow others to concretely assess the robustness of machine learning models and minimize the risk of adversarial attacks.

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$$\min (||\delta||_p) \text{ subject to } f(x + \delta) \leq 0, x + \delta \in [0, 1]^n$$

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Square attack Square attacks were introduced in 2020 as a black-box evasion adversarial attack technique based on querying classification scores without the need for gradient information. As a score based black box attack, this adversarial approach is able to query probability distributions across model output classes, but has no other access to the model itself. According to the paper's authors, the proposed Square Attack required fewer queries than when compared to state-of-the-art score-based black box attacks at the time. To describe the function objective, the attack defines the classifier as $f : [0, 1]^d \rightarrow \mathbb{R}^K$, with d representing the dimensions of the input and K as the total number of output classes. $f_k(x)$ returns the score (or a probability between 0 and 1) that the input x belongs to class k , which allows the classifier's class output for any input x to be defined as $\arg\max_{k=1, \dots, K} f_k(x)$. The goal of this attack is as follows:

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Classifier influence: An attack can influence the classifier by disrupting the classification phase. This may be preceded by an exploration phase to identify vulnerabilities. The attacker's capabilities might be restricted by the presence of data manipulation constraints. Security violation: An attack can supply malicious data that gets classified as legitimate. Malicious data supplied during training can cause legitimate data to be rejected after training. Specificity: A targeted attack attempts to allow a specific intrusion/disruption. Alternatively, an indiscriminate attack creates general mayhem. This taxonomy has been extended into a more comprehensive threat model that allows explicit assumptions about the adversary's goal, knowledge of the attacked system, capability of manipulating the input data/system components, and on attack strategy. This taxonomy has further been extended to include dimensions for defense strategies against

adversarial attacks.

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Adversarial examples An adversarial example refers to specially crafted input that is designed to look "normal" to humans but causes misclassification to a machine learning model. Often, a form of specially designed "noise" is used to elicit the misclassifications. Below are some current techniques for generating adversarial examples in the literature (by no means an exhaustive list).

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Threat modeling – formalize the attackers goals and capabilities with respect to the target system. Attack simulation – formalize the optimization problem the attacker tries to solve according to possible attack strategies. Attack impact evaluation Countermeasure design Noise detection (for evasion-based attack) Information laundering – Alter the information received by adversaries (for model stealing attacks)

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$$\min (|| \delta ||_p) \text{ subject to } C (x + \delta) = t , x + \delta \in [0 , 1]^n$$

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Simple black-box adversarial attacks Simple black-box adversarial attacks is a query-efficient way to attack black-box image classifiers. Take a random orthonormal basis $v_1, v_2, \dots, v_d$ in $\mathbb{R}^d$. The authors suggested the discrete cosine transform of the standard basis (the pixels). For a correctly classified image x , try $x + \epsilon v_1$, $x - \epsilon v_1$, $x + \epsilon v_1, x - \epsilon v_1$, and compare the amount of error in the classifier upon $x + \epsilon v_1, x, x - \epsilon v_1$. Pick the one that causes the largest amount of error.